6. Requirements Specification

Priorities: **MoSCoW** — M must, S should, C could. "Source": where the requirement comes from (proposal objective, literature, or *assumption* to be re-checked in doc 02). Every mandatory requirement has an acceptance criterion that can be tested (doc 09 gives the test IDs).

1. Roles

Driver/resident, Citizen reporter, Item owner/finder (all three are "registered users"), City staff, Administrator. Anonymous visitors may view the public parking map only.

2. Functional requirements

2.1 Parking and reservation (P, R)

ID	Requirement	Pri.	Acceptance criterion	Source	Test
FR-P1	The system shall show parking zones on the map with capacity and current free places	M	For all seeded zones, the popup shows capacity, free places and time of last observation	Obj. 1, A1	T-P1
FR-P2	The system shall show the predicted occupancy 30 minutes ahead for each zone	M	Forecast is shown for every zone; label says "forecast" and model version	Obj. 3, H1	T-P2
FR-P3	Parking data shall be marked as real or synthetic	M	Each zone has `data_source`; UI shows "synthetic data" when applicable	Proposal (risk handling)	T-P3
FR-P4	Staff shall see parking load and forecasts for all zones	S	Staff view lists zones sorted by occupancy with forecast	Obj. 1	T-P4
FR-R1	A registered driver shall reserve a place in a zone for a time slot	M	With free capacity the reservation is saved with status "confirmed"; user receives confirmation	Assumption A2	T-R1
FR-R2	The system shall reject a reservation if capacity for the slot is exhausted, and suggest other zones	M	With a full zone, request is rejected; no overbooking in 50 concurrent requests test	Obj. 2	T-R2
FR-R3	A driver shall cancel own reservation	M	Cancelled reservation frees capacity immediately	Assumption	T-R3
FR-R4	Reservations shall expire after a no-show period	C	Reservation older than limit is set to "expired"	Assumption	T-R4

2.2 Hazard reporting (H)

ID	Requirement	Pri.	Acceptance criterion	Source	Test
FR-H1	A registered user shall submit a report with photo, GPS point (map selection) and text	M	Valid report is stored; report without location is rejected with a message	[11], A3	T-H1
FR-H2	Report text may be in Kazakh, Russian or English	M	Reports in all three languages are accepted and classified	A4	T-H2

ID	Requirement	Pri.	Acceptance criterion	Source	Test
FR-H3	The system shall suggest one of five categories (road damage, lighting, waste, snow and ice, other) with a confidence value	M	On the held-out test set macro-F1 is above the keyword-rule baseline (H1); suggestion and confidence shown in API/UI	Obj. 3, H1	T-H3
FR-H4	Staff shall confirm or correct the category and set the report status (new, in review, resolved, rejected) and verification status	M	Changes are stored with reviewer and time; history visible	[12]	T-H4
FR-H5	The reporter shall see the status of own reports	M	Status shown in "my reports" within 5 s after staff change	[13], A3	T-H5
FR-H6	Reports shall appear on the staff map with filters by category, status, date	M	Filtering returns exactly matching reports	Obj. 1	T-H6
FR-H7	Photos shall be checked so that faces and number plates are not published	S	Public views never show original photos of people/plates; policy documented and tested on test images	[15], NFR-S2	T-H7

2.3 Lost-and-found (L)

ID	Requirement	Pri.	Acceptance criterion	Source	Test
FR-L1	A registered user shall post a lost or found item (category, text, photo, location, time)	M	Valid record saved; ten categories available	[20], A5	T-L1
FR-L2	The system shall return the top-5 candidate matches for a record, using image, text and location	M	On the test pairs, top-5 retrieval rate is above the text-filter baseline (H1)	Obj. 3, H1	T-L2
FR-L3	Users shall filter items by category, date and area	S	Filter returns exactly matching items	Assumption	T-L3
FR-L4	Owner and finder shall confirm or reject a proposed match; status is stored	S	Match status changes to confirmed/rejected; item closes when confirmed	[12]	T-L4
FR-L5	Contact details shall not be public; contact happens through the platform	M	Public API never returns contact data	[15]	T-L5

2.4 Decision-support map for staff (S)

ID	Requirement	Pri.	Acceptance criterion	Source	Test
FR-S1	Staff shall see one map with layers: hazard reports, parking zones, lost-and-found items	M	Layers can be switched on/off independently	A6, Obj. 1	T-S1
FR-S2	The map shall show report density (clusters or heatmap) and counts by category	S	Counts equal database counts for the selected filters	[3]	T-S2
FR-S3	Staff shall export a summary (CSV) of filtered reports	C	CSV contains the filtered rows	Assumption	T-S3
FR-S4	Location data in staff statistics shall be aggregated	M	Statistics show aggregated areas, not exact personal locations	[15], ethics	T-S4

2.5 Common (C)

ID	Requirement	Pri.	Acceptance criterion	Source	Test
FR-C1	Registration, login and role-based access (driver/reporter/finder, staff, administrator)	M	Staff and admin endpoints return 403 for other roles	[15]	T-C1

ID	Requirement	Pri.	Acceptance criterion	Source	Test
FR-C2	One common spatial data model with status and verification fields for sensor data and user reports	M	Schema review: all entities have geometry (where applicable), status, verification fields (doc 07)	Obj. 2	T-C2
FR-C3	Administrator manages parking zones and users	M	Zone create/edit/delete and role change work	Obj. 1	T-C3
FR-C4	Documented REST API (OpenAPI)	S	`/docs` lists all endpoints	doc 05	T-C4
FR-C5	Interface in English; Russian and Kazakh as extension	S	UI strings are externalised; at least two languages in the final version	A4	T-C5

3. Non-functional requirements

ID	Category	Requirement	Pri.	Acceptance criterion	Test
NFR-A1	AI accuracy	Hazard classifier beats keyword rules (macro-F1), lost-item matcher beats text filter (top-5), parking predictor beats historical average (MAE)	M	Results on held-out test sets with bootstrap CI (H1)	T-A1
NFR-A2	AI transparency	Each AI output shows confidence/score and can be overridden by staff or user	M	Inspection of UI/API	T-A2
NFR-P1	Performance	API response time under simulated load	M	Median ≤ 300 ms and 95th percentile ≤ 1 s for list/map endpoints at 50 concurrent users	T-N1
NFR-P2	Performance	AI inference time	S	Hazard classification ≤ 1 s, lost-item matching ≤ 3 s per query on demo hardware	T-N2
NFR-U1	Usability	Mean SUS above the average of 68	M	SUS from the user study	T-U1
NFR-U2	Usability	Task success and time better than baseline workflow	M	Paired test over four tasks (H2)	T-U2
NFR-U3	Transparency	Every status, forecast and AI suggestion shows source/time or confidence	M	Manual inspection	T-U3
NFR-U4	Usability	Responsive layout	S	Usable at 360 px width	T-U4
NFR-S1	Security	Hashed passwords, token expiry, input validation, role checks	M	Tests: no plaintext passwords; injection attempts rejected; 403 tests	T-S5
NFR-S2	Privacy	No faces or number plates in images; aggregated location data; anonymous user-study responses	M	Review checklist and tests	T-S6
NFR-D1	Data	Synthetic data are generated reproducibly and labelled; generator code and seed in the repository	M	Re-running generator gives identical datasets	T-D1
NFR-D2	Data	Hazard labels by two independent annotators; agreement reported (Cohen's kappa)	M	Kappa computed and reported	T-D2
NFR-R1	Reliability	Failure of an AI module must not break the rest of the platform	S	Stop each module; remaining functions work	T-R5

ID	Category	Requirement	Pri.	Acceptance criterion	Test
NFR-M1	Maintainability	Automated tests for backend logic	S	≥ 60 % line coverage on backend	T-M1
NFR-C1	Constraint	Only open-source tools and open/synthetic data; no live deployment; no payments	M	Licence list and review	review

4. Traceability

Objective	Requirements
1. Requirements	FR-P1, FR-H1, FR-L1, FR-S1, doc 02
2. Data model and architecture	FR-C2, FR-R2, NFR-S1, NFR-S2
3. AI modules	FR-P2, FR-H3, FR-L2, NFR-A1, NFR-A2
4. Evaluation	NFR-A1, NFR-P1, NFR-U1, NFR-U2, doc 09

5. Open questions

1. Reservation rules: maximum duration, no-show period, cancellation (decide after interview D1).
2. Exact hazard categories and ontology (confirm after D2/D4).
3. Which authentication method (e-mail, university account).
4. Ethics approval procedure for the user study.